

Brac University
Department of Computer Science and Engineering
Summer 2026

Name:
Student ID:
Section:

10 Marks
35 Minutes

1. If $\beta = 2$, fraction is of 8 bits exponent is of 5 binary bits, What will be the non-negative lowest and highest number that can be generated using the Normalized form of floating point representation? N.B: Exponent range starts from -3.

binary bits for exponent = 5 $\therefore 2^5 = 32$ exponents.

$$e_{\min} = -3 ; e_{\max} = 28$$

$$\text{highest} = + 0.11111111 \times 2^{28} = 267911168$$

$$\text{non-neg lowest} = + 0.100000000 \times 2^{-3} = 0.0625$$

2. Derive the formula for machine epsilon (ϵ_M) for the Standard form of floating point representation.

Same as lecture notes

3. For a system if $\beta = 2$, $m=4$ and $e \in \{-3,3\}$ then how many non-negative number can be represented in a system following Normalized form?

$$e \in \{-3,3\} = 7 \text{ exponents}$$

$$\text{total non-negatives} = 2^4 \times 7 = 112$$

4. If $x=3/8$, and $y=5/8$ find $fl(x*y)$ where $m=2$, $e \in \{-3,3\}$ and the system follows Normalized Form of floating point representation. Also find the relative rounding error.

$$x \times y = \frac{3}{8} \times \frac{5}{8} = \frac{15}{64} = 0.001111 = 0.1111 \times 2^{-2}$$

for $m=2$

$$0.1111 \times 2^{-2} \quad 0.1111 \times 2^{-2} \quad 0.100 \times 2^{-1}$$

$$= (2^{-1} + 2^{-2} + 2^{-3}) \times 2^{-2} = 2^{-1} \times 2^{-1}$$

$$= 0.21875 = 0.25$$

$$\text{avg} = \frac{0.21875 + 0.25}{2}$$

$$= 0.234375$$

$$= 0.1111 \times 2^{-2}$$

$$\therefore fl(x \times y) = 0.100 \times 2^{-1} = 0.25$$

$$\text{relative rounding error} = \frac{|0.25 - 0.234375|}{|0.234375|} = 0.0667$$

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1. If $\beta = 2$, fraction is of 4 bits exponent is of 3 binary bits, What will be the non-negative lowest and highest number that can be generated using the Normalized form of floating point representation? N.B: Exponent range starts from -1.

binary bits for exponent = 3 $\therefore 2^3 = 8$ exponents

$e_{\min} = -1$; $e_{\max} = 6$

highest = $+0.11111 \times 2^6 = 62$

non-neg lowest = $+0.10000 \times 2^{-1} = 0.25$

2. Derive the formula for machine epsilon (ϵ_M) for the Normalized form of floating point representation.

Same as lecture notes

3. For a system if $\beta = 2$, $m=4$ and $e \in \{-3,4\}$ then how many non-negative number can be represented in a system following Denormalized form?

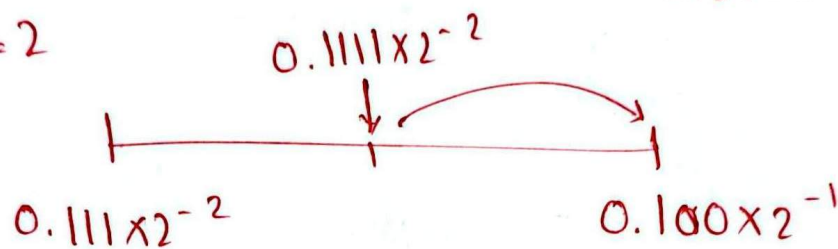
$$e \in \{-3,4\} = 8 \text{ exponents}$$

$$\text{total non-negatives} = 2^4 \times 8 = 112$$

4. If $x=3/8$, and $y=5/8$ find $fl(x*y)$ where $m=2$, $e \in \{-3,3\}$ and the system follows Normalized Form of floating point representation. Also find the relative rounding error.

$$x \times y = \frac{3}{8} \times \frac{5}{8} = \frac{15}{64} = 0.001111 = \cancel{0.1111 \times 2^{-2}} = 0.1111 \times 2^{-2}$$

for $m=2$



$$\begin{aligned} \text{for } m=2 &= (2^{-1} + 2^{-2} + 2^{-3}) \times 2^{-2} = 2^{-1} \times 2^{-1} \\ &= 0.21875 = 0.25 \end{aligned}$$

$$\text{avg} = \frac{0.21875 + 0.25}{2}$$

$$= 0.234375$$

$$= 0.1111 \times 2^{-2}$$

$$\therefore fl(xy) = 0.100 \times 2^{-1} = 0.25$$

$$\text{relative rounding error} = \frac{|0.25 - 0.234375|}{|0.234375|} = 0.0667$$

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1. If $\beta = 2$, fraction is of 5 bits exponent is of 4 binary bits, What will be the non-negative lowest and highest number that can be generated using the Normalized form of floating point representation? N.B: Exponent range starts from -2.

binary bits for exponent = 4 $\therefore 2^4$ exponents = 16 exponents
 $e_{\min} = -2, e_{\max} = 13$
highest = $+0.111111 \times 2^{13} = 8064$
non-negative lowest = $+0.100000 \times 2^{-2} = 0.125$

2. Derive the formula for machine epsilon (ϵ_M) for the Denormalized form of floating point representation.

Same as lecture notes

3. For a system if $\beta = 2$, $m=4$ and $e \in \{-3,3\}$ then how many non-negative number can be represented in a system following Normalized form?

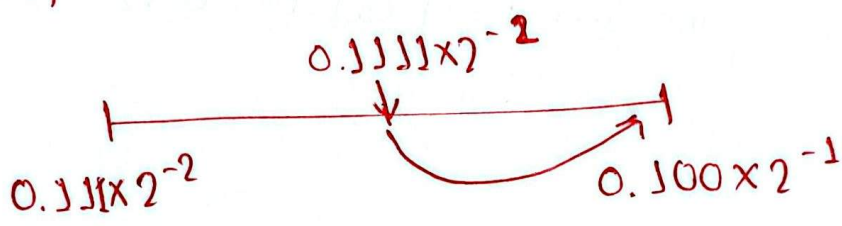
$$e \in \{-3,3\} = 7 \text{ exponents.}$$

$$\text{total non-negative numbers} = 2^4 \times 7 = 112$$

4. If $x=3/8$, and $y=5/8$ find $fl(x*y)$ where $m=2$, $e \in \{-3,3\}$ and the system follows Normalized Form of floating point representation. Also find the relative rounding error.

$$x \times y = \frac{3}{8} \times \frac{5}{8} = \frac{15}{64} = 0.001111 = 0.1111 \times 2^{-2}$$

for $m=2$,



$$\begin{aligned} & 0.111 \times 2^{-2} & 0.1111 \times 2^{-2} & 0.100 \times 2^{-1} \\ & \downarrow & \downarrow & \uparrow \\ & (2^{-1} + 2^{-2} + 2^{-3}) \times 2^{-2} & & 2^{-1} \times 2^{-1} \\ & = 0.21875 & & = 0.25 \end{aligned}$$

$$\text{avg} = \frac{0.21875 + 0.25}{2}$$

$$= 0.234375$$

$$= 0.1111 \times 2^{-2}$$

$$\therefore fl(x \times y) = 0.100 \times 2^{-1} = 0.25$$

$$\text{relative rounding error} = \frac{|0.25 - 0.234375|}{|0.234375|} = 0.0667$$